

Python Phase 1 – Cheat Sheet (Cloud Security Foundation)

FUNDAMENTALS

- Variables, Data Types, Loops, Conditionals
- Functions, Modular Code, Return Values
- Lists, Tuples, Dictionaries, Sets
- String Manipulation, Slicing, Formatting
- Exception Handling (try/except)

FILE HANDLING (SECURITY-RELEVANT)

- Open/Read/Write Files
- CSV Handling (csv.reader, csv.writer)
- JSON Handling (json.dumps, json.loads)
- Directory Automation with os & shutil

AUTOMATION & SYSTEM INTERACTION

- Automating repetitive tasks
- Pattern-based automation
- Clipboard automation (pyperclip)
- Bulk file operations & renaming

PYTHON + CLOUD/SECURITY USE CASES

- Log parsing (CSV, JSON)
- API data handling (weather API)
- Directory scanning & automation
- File monitoring scripts

MINI PROJECTS SUMMARY

1. Current Weather Data Fetch

- Uses API, JSON parsing, error handling

2. Multi Clipboard Program

- Uses shelve, persistent storage, automation

3. Quiz File Generator

- Randomization, file writing, dictionaries

4. Remove CSV Header Script

- os module automation, CSV parsing

5. Coin Flip Streak Simulation

- Loops, simulation, random module

6. Collatz Sequence

- Mathematical logic, recursion/iteration

CORE MODULES USED

- os — directory automation
- shutil — file operations
- csv — log & dataset processing
- json — API data and configs
- random — randomness & simulations
- shelve — persistent storage

SECURITY MAPPING

- File automation → log analysis
- CSV/JSON parsing → cloud logs, API responses
- Automation scripts → incident response helpers
- Randomness → testing & fuzzing basics
- Exception handling → stable security scripts

TIPS

- Always format and validate inputs
- Keep code modular (functions)

- Comment your logic for GitHub readers
- Practice JSON & CSV — critical for cloud logs